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PI Name:	Raber, Jacob Ph.D.		
Project Title:	The Relation Between Cognitive Injury, Network Stability, and Epigenetic Change Following Exposure to Space Radiation		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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PI Email:	raberj@ohsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	503-494-1524
Organization Name:	Oregon Health & Science University		
PI Address 1:	Behavioral Neuroscience and Neurology		
PI Address 2:	L470, 3181 SW Sam Jackson Park Road		
PI Web Page:			
City:	Portland	**State:**	OR
Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
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Key Personnel Changes/Previous PI:	Oct 2015: Tessa Marzulla went back to school and Sydney Weber took over her duties.		
COI Name (Institution):	Impey, Soren (Oregon Health & Science University) Turker, Mitchell (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
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Task Description:	Space radiation poses a significant hazard to astronaut health both during the mission and at later times. However, uncertainty exists with respect to the level of risk to the CNS (central nervous system) following radiation doses relevant to those encountered during space missions. Radiation exposure in the brain affects the hippocampus, a structure critical for cognition. This proposal addresses section 1.2 of NASA Research Announcement NNJ12ZSA001N entitled, “Risk of Acute or Late Central Nervous System Effects from Space Radiation.” The proposed work focuses on changes in hippocampal network stability, including epigenetic mechanisms, in the development of cognitive dysfunction after space irradiation. Radiation exposure of the CNS significantly affects the hippocampus. Our preliminary data show impaired cognition in mice at two weeks following low dose space radiation. During spatial exploration, when an animal enters a specific location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory and resulting in the transcription of the immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc). Thus, analyzing Arc provides a reliable method to detect the number of active cells forming environment-specific maps and to capture the environmental specificity of place fields. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen across visits in different environments. The proposed work will focus initially on effects of space radiation on the stability of hippocampal networks, assessed by analyzing the expression of Arc and other genes important for memory formation. We hypothesize that the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability. Promoter region DNA methylation (5mC) of Arc leads to decreased expression and space radiation induces changes in DNA methylation. Moreover, DNA methylation plays a key role in regulating expression of genes required for spatial learning and memory. Thus, study of Arc expression and DNA methylation after charged particle exposure is warranted, followed by a more global search for DNA methylation changes in genes involved in learning and memory. A second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC), is abundant in the brain and found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. In this proposal, we will test the hypotheses that the detrimental effects of space radiation on cognition are associated with Arc-related changes in the hippocampal networks involved in learning and memory and that these changes are related to space radiation-induced perturbation of 5mC and/or 5hmC. Our approach will link assessment of hippocampus-dependent cognitive changes, hippocampal network stability, and alterations in global DNA methylation and expression of genes involved in learning and memory. In Specific Aim 1, we will test the hypothesis that low-dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on cognitive performance in object recognition and spatial learning and memory requiring navigation. In Specific Aim 2, we will test the hypothesis that low dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on hippocampal networks involving Arc, and that cognitive injury following space irradiation is associated with changes in hippocampal network stability. In Specific Aim 3, we will test the hypothesis that cognitive changes following space irradiation are associated with alterations in DNA methylation and expression of Arc and other genes involved in learning and memory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humans are exposed to ionizing irradiation under a variety of conditions associated with medical treatment. While the risks of developing cognitive impairments in those situations are fairly well established, uncertainty exists with respect to other types of radiation exposure, such as those experienced in the space environment. There is a critical need to establish the level of risk to the CNS following radiation doses relevant to those encountered during space missions. NASA has identified four categories of radiation risk in the Integrated Research Plan (IRP) and their associated research and technology gaps. This project is pertinent to 1.2 “Risk of Acute or Late Central Nervous System Effects from Space Radiation” and to Space Radiation Risks CNS1, CNS2, CNS6, Degenerative 1, and Degenerative 2. The project focuses on the role of changes in hippocampal network stability and epigenetic mechanisms involved in these changes in the development of cognitive dysfunction after space irradiation. CNS-related topics expressly targeted by the IRP and that are specifically or indirectly addressed in the current project include early and late effects of space irradiation on cognitive function and cellular and molecular biomarkers that might be associated with cognitive injury. Innovative technologies are being developed for integrated functional assays to assess effects of space radiation on brain function.
	This project focuses on changes in hippocampal network stability and epigenetic mechanisms involved in the development of cognitive dysfunction after space irradiation. Radiation exposure of the central nervous system (CNS) can significantly affect the hippocampus, a structure critical for cognitive function. During spatial exploration, when an animal enters a specific location a distinct cell responds (known as the cell's place field). As a result of these place fields, hippocampal pyramidal neurons in the CA areas display robust high frequency activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory. Following visits to an environment for a second time, the same cells form stable neural networks and respond as a result of cellular activity induced by a learning experience. While place fields are highly reproducible and stable across multiple visits to the same environment, this stability is not seen in when different environments are visited. The detrimental effects of charged particles on hippocampus-dependent cognition are likely associated with changes in hippocampal networks. Analysis of the precise temporal kinetics of the plasticity-related immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc) using a method called catFISH (cellular compartment analysis of temporal activity using fluorescence in situ hybridization) provides an ideal and reliable approach for studying network stability. In a prior study we showed that Arc protein expression in the hippocampus is sensitive to brain-only exposure to 56Fe ions (600 MeV, 1 Gy). Arc expression is reduced with aging in the CA1 “place cells” within the mouse hippocampus and this reduction is correlated with increased promoter region DNA methylation (5mC). This is not surprising because changes in DNA methylation play a key role in regulating expression of genes required for spatial learning and memory. Gamma and space irradiation, including 56Fe ions and protons, induce changes in DNA methylation. Of significance for this project, recent studies show that a second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC) is abundant and stable in the brain. The TET (ten eleven translocation) enzymes convert 5mC to 5hmC, which likely play a role in gene regulation because 5hmC is often found at promoters of active genes. In contrast, 5mC is found at the promoters of inactive genes. Changes in 5mC, which are induced by environmental exposures including radiation and behavioral modification, play a role in processes including cancer and neurological disease. A similar role for 5hMC can be assumed due to its abundance in the brain, though 5hmC changes after radiation exposure and in neurological disease has not yet been elucidated. We hypothesize that the detrimental effects of charged particles on hippocampus-dependent cognition are associated with Arc-related changes in the hippocampal networks involved in synaptic plasticity and

Task Progress:	memory. We further hypothesize that these changes are related to perturbation of 5mC and/or 5hMC in the hippocampus that affect Arc expression and expression of other learning and memory genes in the same signaling pathways and networks. Our data show profound impairments in cognitive performance in 6-month-old C57BL/6J wild-type (Wt) mice two weeks following irradiation with 56Fe, 28Si, and proton irradiation. Our data also show that radiation-induced cognitive injury is associated with cytosine methylation (5mC) and hydroxymethylation (5hmC) of cytosine in the hippocampus. Regions with decreased 5mC were associated with pathways linked to neuronal differentiation, process growth, axogenesis, and synaptic function. Regions with both decreased and increased 5hmC had highly significant overlap with pathways linked to decreased 5mC. These results are consistent with coupled DNA methylation-oxidation (Tet-mediated oxidation) and active demethylation. Consistent with the dose-dependent pattern of the data, the mice irradiated with one dose of 56Fe were cognitive impaired, while those irradiated with another dose were not. We also observed the relevance of the altered pathways to neurodegenerative conditions like Parkinson disease and Alzheimer's disease. The brain is unique among differentiated tissues because many brain regions have high levels of 5hmC that persist throughout development. In particular, up to 20% of modified CpGs have been proposed to be 5hmC in hippocampus. Our data indicate that significant increases in 5hmC in the hippocampus is most significantly correlated (spatially) with regions that have increased 5mC. The co-localization of upregulated DMRs and DHRs is consistent with previous observations suggesting that 5hmC is an abundant and persistent mark in brain. Moreover, it raises the possibility that deposition of hippocampal 5mC and 5hmC may be mechanistically coupled. We also observed for the first time tissue-dependent changes in DNA methylation profiles in response to proton irradiation that reflect perturbations in tissue-specific pathways (e.g., neuronal specific pathways in the hippocampus and vascular development pathways in the ventricle). We compared DNA methylation (5mC and 5hmC) in the hippocampus and left ventricle of 11-month-old Wt mice twenty weeks following proton irradiation at 6 months. The gene region-specific methylation profiles in both tissues revealed significant major pathway changes that are related to known pathophysiologic processes. Gene pathway analyses of domains that showed decreased 5mC in response to proton radiation revealed striking enrichment for key tissue-specific pathways, suggesting that epigenetic remodeling was associated with cellular differentiation responses. Moreover, these pathways were highly enriched for genes that are key determinants of cell-fate identity in brain and heart. Overlap between gene pathway data in the ventricle and hippocampus was also observed, suggesting that proton radiation elicits both a tissue-specific and a general radiation response. Regions with decreased 5hmC in the left ventricle were enriched for muscle and heart-specific gene pathways, while hippocampal tissue showed enrichment for genes linked to small G-protein signaling and associated cytoskeletal remodeling. In summary, our novel data demonstrate the power of this approach and by extension that the proposed outcome measures are sensitive and specific to detect effects of relatively low dose irradiation within a few weeks after radiation exposure and also indicate the anatomical specificity and networks involved in these effects.

Bibliography Type:	Description: (Last Updated: 10/30/2023)
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